Sharukesh M

B.Tech AI & Data Science Student

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Profile Summary

- Third-Year AI & Data Science student skilled in deploying scalable ML/real-time systems using Python, Java, and C. Experienced in deep learning, Big data, and API development, with a strong aptitude for high-performance computing and optimization. Eager to apply analytical problem-solving in a collaborative, innovative team.

Education

B.Tech in Artificial Intelligence & Data Science **CGPA: 8.78 (Current)**
Amrita School of AI, Amrita Vishwa Vidyapeetham, Coimbatore

High school & Higher secondary
Sri Chaitanya techno school

Skills

Programming Languages: Python, C/C++, Java, Scala, Matlab, SQL
AI/ML Libraries: PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Anaconda, Hugging Face, LangChain
Tools & Platforms: Docker & k8s, API Development, Linux/Ubuntu Environment, HPC & Optimization
Core Competencies: Analytical Problem-Solving, Collaborative Teamwork
Areas of interest: ML/DL, HPC, Big Data

Internship Experience

AI Intern — Telesto Energy **December 2025 – March 2026**
- Built a scalable knowledge base using the RAG-Anything framework and AWS Bedrock for LLM orchestration.
- Integrated Redis, FAISS for high-speed vector search, and Neo4j to enable graph-based multi-hop retrieval (Graph RAG).
- Developed and deployed an OpenCV and FastAPI-based document restoration pipeline to enhance degraded archival records.
- Improved OCR accuracy and digitization quality by removing brown artifacts, normalizing backgrounds, and sharpening text.

AI Intern — Discovery AI LLC **September 2025 – November 2025**
- Developed an interactive virtual rug visualizer for live, in-room product previews.
- Engineered a CV pipeline using SAM (segmentation) and Midas (depth) for realistic rug placement.
- Deployed real-time, client-side image processing and manipulation using Pyodide and OpenCV.
- Designed an end-to-end workflow from image capture to 3D model overlay in complex environments.

AI Application Intern — Incept AI **March 2025 – April 2025**
- Developed and deployed Microsoft Security Copilot plugins via CI/CD pipelines to enhance security workflows.
- Engineered RAG pipelines (DeepSeek R1) and agentic workflows for dynamic, context-aware KQL generation.
- Improved SOC efficiency by automating query generation and reducing analyst response times.

Projects

Skin Lesion Classification **January 2025 – April 2025**
- Designed a multi-task CNN (U-Net + ResNet) for accurate skin lesion segmentation and classification (RCCN).
- Built a secure web/mobile app (FastAPI, PostgreSQL) for diagnosis, treatment recommendations, and geolocation-based appointment booking.

Scalable Academic Research Engine **August 2025 – October 2025**

- Engineered a scalable data pipeline using Apache Spark (Scala) and HDFS to process and generate SciBERT embeddings for a massive corpus of academic papers.
- Deployed a high-throughput, CUDA-accelerated vector search backend using FAISS for real-time similarity search.

Hemo-Flash: Efficient Hematology Detection **January 2026 – March 2026**

- Developed **Hemo-Flash**, a lightweight computer vision system based on a custom **YOLOv11** architecture for real-time detection and classification of RBCs, WBCs, and platelets.
- Achieved a **86% reduction in model parameters** compared to SOTA YOLO models while maintaining equivalent diagnostic accuracy, significantly optimizing the model for resource-constrained environments.
- Engineered a high-performance inference pipeline for automated blood cell counting, balancing computational efficiency with State-of-the-Art (SOTA) precision.

Publications

- **Analysis of Deep and Graph Neural Networks for Enhanced DDoS Attack Detection: A Pathway to Hybrid Models**
  7th International Conference on Advancements in Computing (ICAC), IEEE Xplore **2025**
  Co-authored with S Athif, SR Nidish, VS Vignesh, SS Amrita, D Sasi
- **Design and Implementation of an Autonomous Self-Controlled Delivery Robot with an OTP-Based Secure Delivery**
  1st International Conference on Smart and Intelligent Systems (SISCON) **2025**
  Co-authored with S Devarajan, S Athif, G Vinod, S Abhishek
- **An Intelligent Plug-In Playable Orchestration Model that Combines Gen-AI Prompting, RAG and Agentic AI for Efficient Cybersecurity Operations**
  International Conference on Smart Innovation, Systems and Technologies **2025**
  Co-authored with S Sundarakrishna, VS Vignesh, P Menon Perinchery